Medvale Internal Medicine
Chapter 2F: Pulmonary Vascular Disease

Pulmonary Embolism, DVT, Pulmonary Hypertension, and Cor Pulmonale

Originality note.
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Why this matters clinically

Pulmonary vascular disease is where lung physiology, venous thrombosis, right-heart mechanics, and inpatient decision-making meet. A patient with pulmonary embolism may appear with pleuritic chest pain, unexplained tachycardia, syncope, isolated hypoxemia, hemoptysis, or shock. A patient with chronic pulmonary hypertension may present more slowly with exertional dyspnea, loud P2, presyncope, edema, abdominal fullness, or progressive exercise intolerance. In both settings, the central clinical question is not simply whether pulmonary artery pressure is high; it is whether right ventricular output can keep up with pulmonary vascular resistance.

For exam and ward reasoning, pulmonary vascular disease is high yield because the initial tests depend on pretest probability and hemodynamic stability. A stable patient with low probability may need PERC or D-dimer logic; an unstable patient with suspected massive pulmonary embolism may need immediate bedside echo, empiric anticoagulation when safe, and reperfusion consideration before a perfect diagnostic sequence is available.

Core illness scripts

Syndrome	Typical story	Key diagnostic anchor	Immediate danger
Acute pulmonary embolism	Abrupt dyspnea, pleuritic chest pain, tachycardia, hypoxemia, syncope, hemoptysis, or unexplained respiratory alkalosis after risk exposure such as surgery, immobility, malignancy, pregnancy/postpartum state, estrogen therapy, thrombophilia, or prior VTE.	Clinical probability plus D-dimer or imaging. CTPA is common first-line imaging in stable patients; V/Q scan helps when CTPA is undesirable.	Right ventricular strain, obstructive shock, cardiac arrest, recurrent embolization.
Deep venous thrombosis	Unilateral leg swelling, pain, warmth, erythema, or asymmetry; may be silent. Proximal clot is more likely to embolize than isolated distal calf clot.	Compression ultrasonography; repeat imaging can be used when suspicion remains high after an initial negative proximal study.	PE, post-thrombotic syndrome, anticoagulation bleeding complications.
Pulmonary arterial hypertension	Progressive exertional dyspnea, fatigue, chest pressure, presyncope or syncope, loud P2, RV heave, tricuspid regurgitation murmur, edema.	Echocardiography screens; right heart catheterization confirms hemodynamics and classifies severity.	Progressive RV failure and sudden decompensation.
Cor pulmonale	Chronic lung disease or hypoxemia with right-sided congestion: JVD, hepatomegaly, ascites, edema, loud P2, RV lift. COPD, ILD, OSA/OHS, and chronic thromboembolic disease are common mechanisms.	Evidence of lung disease plus RV enlargement/dysfunction and pulmonary hypertension physiology.	Volume-sensitive RV failure, worsening gas exchange, arrhythmias, renal/hepatic congestion.

Pathophysiology: from clot to right-heart failure

A pulmonary embolus raises pulmonary vascular resistance by physically blocking arterial flow and by triggering vasoconstriction through hypoxemia, platelet mediators, and neurohumoral responses. The right ventricle is adapted to a low-pressure circuit. A sudden afterload rise can dilate the right ventricle, stretch the tricuspid annulus, reduce RV systolic output, and shift the interventricular septum toward the left ventricle. That septal shift reduces left ventricular filling, lowering systemic blood pressure. This is why massive PE is an obstructive shock state rather than a primary lung-only problem.

Gas exchange in PE is variable. Many patients develop an increased alveolar-arterial oxygen gradient because ventilated alveoli are underperfused and because regional atelectasis or low mixed venous oxygen content worsens arterial oxygenation. A normal oxygen saturation does not exclude PE. Hyperventilation can produce low PaCO2 early. In suspected PE, hypotension, syncope, rising lactate, high troponin, elevated BNP, or echo evidence of RV strain suggests a clot burden or physiologic response large enough to threaten circulation.

Original mechanism diagram: acute PE as an RV afterload problem

Venous thrombus forms -> **Embolus reaches pulmonary artery** -> **Pulmonary vascular resistance rises**

Right ventricle dilates and strains -> **LV preload falls** -> **Hypotension or shock**

Diagnostic approach to suspected pulmonary embolism

The first branch point is stability. If the patient is hypotensive, peri-arrest, or has severe persistent shock physiology, the diagnostic pathway becomes compressed. Bedside echocardiography, lower-extremity ultrasound, ECG, chest radiograph, ABG/VBG, lactate, and troponin/BNP can support the diagnosis while clinicians prepare anticoagulation or reperfusion if bleeding risk allows. If the patient is stable, the goal is to avoid both missed PE and unnecessary CT pulmonary angiography. That requires explicit pretest probability.

Pretest probability tools do not diagnose PE; they organize uncertainty. The Wells score assigns points to clinical DVT signs, PE as the most likely diagnosis, heart rate over 100/min, recent surgery or immobilization, prior VTE, hemoptysis, and malignancy. A low-probability patient who meets all PERC criteria may need no D-dimer. PERC is intended only when the clinician already believes the patient is low risk. It includes age under 50 years, heart rate under 100/min, oxygen saturation at least 95%, no hemoptysis, no estrogen use, no prior VTE, no unilateral leg swelling, and no recent surgery or trauma. If probability is not low or PERC is not satisfied, D-dimer or imaging follows depending on clinical probability.

Clinical state	Reasonable next step	Interpretation
Unstable patient with high suspicion	Resuscitate, bedside echo for RV strain, consider CTPA only if safe. Start anticoagulation if no major contraindication; consider systemic thrombolysis, catheter-directed therapy, or embolectomy for high-risk PE.	Do not delay life-saving therapy for a perfect test when PE is causing obstructive shock.
Low probability and PERC negative	No D-dimer and no imaging in the correct clinical context.	This is a rule-out strategy for very low-risk patients, not a universal screening checklist.

Clinical state	Reasonable next step	Interpretation
Low/intermediate probability but PERC positive or not applicable	High-sensitivity D-dimer. Age-adjusted D-dimer may be used in patients over 50 in many pathways.	Negative D-dimer can rule out PE when pretest probability is not high. Positive D-dimer is nonspecific.
High probability or positive D-dimer	CT pulmonary angiography when renal function/contrast tolerance and pregnancy considerations allow. V/Q scan if CTPA is undesirable and chest radiograph is suitable.	A negative D-dimer should not be used to dismiss high-probability PE.

Numeric anchors for graph RAG.
Heart rate over 100/min appears in Wells reasoning and PERC exclusion. Oxygen saturation below 95% fails PERC. Age-adjusted D-dimer is commonly calculated as age x 10 ng/mL FEU for patients older than 50 when FEU units are used. Therapeutic anticoagulation for acute VTE generally has a minimum treatment phase of 3 months unless bleeding risk, contraindication, or goals-of-care issues dominate.

Imaging and laboratory interpretation

CT pulmonary angiography directly visualizes intraluminal filling defects and can identify alternate diagnoses such as pneumonia, pneumothorax, aortic disease, or malignancy. V/Q scanning is useful when contrast should be avoided, especially in patients with severe contrast allergy, significant renal limitations, or pregnancy-specific imaging pathways. A normal chest radiograph makes V/Q interpretation more useful; extensive underlying lung disease can make the scan nondiagnostic. Lower-extremity compression ultrasound may establish VTE when PE imaging is delayed or unsafe; finding a proximal DVT in a patient with PE symptoms often justifies treatment even if chest imaging is not immediately available.

ECG findings are neither sensitive nor specific. Sinus tachycardia is common. Right heart strain patterns, right axis deviation, T-wave inversions in V1-V4, incomplete or complete right bundle branch block, and the classic S1Q3T3 pattern can appear but should not be required. Troponin and BNP are risk markers rather than diagnostic tests. Elevation suggests RV myocardial stretch or injury and helps identify intermediate-risk PE when blood pressure is preserved.

Risk stratification of confirmed PE

After PE is diagnosed, the next question is physiologic risk. High-risk PE is defined by hemodynamic instability such as sustained hypotension, shock, or cardiac arrest attributable to PE. Intermediate-risk PE has preserved systemic pressure but evidence of RV dysfunction, myocardial injury, or clinical severity. Low-risk PE lacks these high-risk features and may be appropriate for early discharge in selected patients with reliable follow-up, low bleeding risk, and adequate home support.

Risk category	Typical findings	Management logic
High-risk / massive PE	Hypotension, shock, need for vasopressors, cardiac arrest, severe RV failure.	Immediate anticoagulation if possible plus reperfusion consideration: systemic thrombolysis, catheter-directed intervention, surgical embolectomy, or ECMO bridge in selected centers.
Intermediate-risk / submassive PE	Normotensive but RV strain on echo/CT, positive troponin or BNP, high clinical severity, or worsening oxygenation.	Anticoagulation and close monitoring. Rescue thrombolysis or catheter therapy if decompensation occurs. Routine thrombolysis is limited by bleeding risk.
Low-risk PE	Stable vitals, no RV strain, no biomarker injury, low clinical severity, low bleeding risk.	Anticoagulation. Selected patients may be outpatient candidates when social and follow-up criteria are strong.

Management of acute PE and DVT

Anticoagulation prevents clot extension and recurrent embolization while the body organizes and lyses thrombus. It does not instantly dissolve most clot. For stable patients without contraindications, direct oral anticoagulants are commonly used because they avoid routine INR monitoring and have predictable dosing. Unfractionated heparin is favored when rapid reversal may be needed, when procedures are likely, in severe renal failure, or in unstable patients who may need thrombolysis or embolectomy. Low-molecular-weight heparin remains useful in pregnancy and selected malignancy-associated thrombosis scenarios. Warfarin requires bridging at initiation for acute VTE because early vitamin K antagonist therapy can transiently reduce protein C and S before full anticoagulant effect.

Therapy	When it fits	Important limitations
DOACs such as apixaban or rivaroxaban	Many stable nonpregnant adults with adequate renal/hepatic function and no major drug interactions.	Avoid or individualize in pregnancy, severe renal disease, high-risk antiphospholipid syndrome, major interacting medications, and selected extremes of body size or absorption issues.
Unfractionated heparin infusion	Hemodynamic instability, high bleeding concern with need for rapid stop/reversal, severe renal impairment, or likely procedure.	Requires aPTT or anti-Xa monitoring; HIT risk; hospitalization.
LMWH	Pregnancy-associated VTE, selected cancer-associated VTE, predictable parenteral therapy.	Renal dose adjustment; injections; cost; monitoring in special populations.
Warfarin	Mechanical valves, antiphospholipid syndrome in many cases, cost/access limitations, severe renal disease when DOACs are unsuitable.	Requires INR monitoring, food/drug interactions, bridging for acute VTE.
IVC filter	Acute proximal DVT/PE with absolute contraindication to anticoagulation, or recurrent PE despite adequate anticoagulation in selected cases.	Does not treat existing clot and can increase DVT/filter thrombosis risk; retrieve when no longer needed.

Duration is a recurrence-versus-bleeding decision. A transient major provoking factor such as surgery often supports a finite 3-month treatment phase. Unprovoked proximal DVT or PE, persistent risk factors, active cancer, recurrent VTE, or chronic thromboembolic disease often push toward extended therapy if bleeding risk is acceptable. Distal isolated DVT may be anticoagulated or monitored with serial ultrasound depending on symptoms, extension risk, and patient factors. The exam trap is to treat every clot forever or to stop all anticoagulation at discharge.

Scenario	Typical duration logic	Reasoning
VTE provoked by major transient risk factor	Usually 3 months.	Risk falls after the trigger resolves; extended therapy may add bleeding without enough recurrence benefit.
Unprovoked proximal DVT or PE	At least 3 months, then reassess for extended therapy.	Recurrence risk persists after stopping anticoagulation.
Active cancer-associated VTE	Extended therapy while cancer is active or treatment continues, if bleeding risk permits.	Malignancy is a persistent prothrombotic state.
Pregnancy-associated VTE	Therapy through pregnancy and postpartum; minimum total duration is commonly at least 3 months with at least 6 weeks postpartum coverage.	Pregnancy and early postpartum are high-risk hypercoagulable periods.
Recurrent unprovoked VTE	Often indefinite unless bleeding risk is prohibitive.	History of recurrence strongly raises future recurrence risk.

DVT: the venous source and its own clinical problem

Most clinically important PE begins as thrombus in the deep veins of the legs or pelvis. Virchow triad organizes risk: venous stasis, endothelial injury, and hypercoagulability. Stasis occurs with immobilization, paralysis, long hospitalization, or heart failure. Endothelial injury follows surgery, trauma, central venous instrumentation, or inflammation. Hypercoagulability may be acquired, as in malignancy, estrogen exposure, pregnancy, nephrotic syndrome, antiphospholipid syndrome, or heparin-induced thrombocytopenia; it may also be inherited, as in factor V Leiden or prothrombin gene mutation.

The physical exam cannot reliably exclude DVT. Calf tenderness, edema, warmth, erythema, and asymmetry raise suspicion, but many thrombi are silent. Compression ultrasonography is the central test. A noncompressible femoral or popliteal vein is diagnostic. If suspicion remains high after a negative proximal ultrasound, repeat imaging in about 5 to 7 days is often used to detect extension from calf veins into proximal veins. D-dimer is useful only when pretest probability is low enough for a negative test to rule out disease.

Pulmonary hypertension: classification and confirmation

Pulmonary hypertension is not a single disease. It is a hemodynamic finding that requires classification because therapy differs dramatically by cause. The current hemodynamic definition uses mean pulmonary arterial pressure greater than 20 mm Hg measured by right heart catheterization. Pulmonary arterial hypertension also requires elevated pulmonary vascular resistance and a pulmonary arterial wedge pressure that does not indicate left-sided filling pressure as the primary driver. Echocardiography estimates probability, but right heart catheterization confirms diagnosis, quantifies pressure and resistance, separates pre-capillary from post-capillary physiology, and guides treatment.

WHO group	Dominant mechanism	Examples	Treatment principle
Group 1: Pulmonary arterial hypertension	Pulmonary arteriolar remodeling and high pulmonary vascular resistance.	Idiopathic, heritable, connective tissue disease, congenital heart disease, portal hypertension, HIV, drug/toxin associated.	Specialized PAH therapy after expert evaluation; vasoreactivity testing in selected idiopathic/heritable/drug-associated disease.
Group 2: Left heart disease	Back pressure from left atrial/ventricular disease.	HFpEF, HFrEF, valvular disease, cardiomyopathy.	Treat left heart disease and congestion; PAH vasodilators can harm selected patients.
Group 3: Lung disease/hypoxia	Hypoxic vasoconstriction, vascular bed loss, parenchymal destruction.	COPD, ILD, OSA, obesity hypoventilation, high altitude.	Treat lung disease, hypoxemia, sleep-disordered breathing; oxygen when indicated.
Group 4: Pulmonary artery obstruction	Chronic organized thromboembolic obstruction.	CTEPH after PE; chronic thrombotic or embolic disease.	Lifelong anticoagulation; evaluate for pulmonary endarterectomy, balloon angioplasty, and targeted therapy.
Group 5: Multifactorial/unclear	Mixed mechanisms.	Sarcoidosis, hematologic disorders, chronic kidney disease, metabolic disorders.	Treat underlying disorder and refer when severe or unexplained.

Symptoms reflect inability to increase cardiac output during exertion. Early disease may have normal resting oxygen saturation and a subtle exam. As pulmonary vascular resistance rises, the right ventricle hypertrophies and later dilates. Classic findings include a loud pulmonary component of the second heart sound, right ventricular heave, tricuspid regurgitation murmur, jugular venous distention, hepatomegaly, ascites, and peripheral edema. Syncope, especially with exertion, is a high-risk clue because it suggests fixed output or arrhythmia during increased demand.

Original diagnostic map: pulmonary hypertension

Echo suggests elevated RV pressure	->	**Search for cause**	->	**Right heart catheterization confirms**
Pre-capillary?	->	**Post-capillary?**	->	**Classify WHO group and treat cause**

Chronic thromboembolic pulmonary hypertension

CTEPH is an important cause of potentially treatable pulmonary hypertension. It occurs when thromboembolic material fails to resolve and becomes organized within pulmonary arteries, producing chronic obstruction and secondary small-vessel remodeling. Symptoms may appear months to years after acute PE, but some patients do not recall a prior embolic event. Persistent dyspnea after PE should not be dismissed as deconditioning without reassessment. V/Q scanning is highly useful as a screening test for chronic thromboembolic disease; right heart catheterization and pulmonary angiographic imaging define hemodynamics and operability. Pulmonary endarterectomy can be curative in suitable proximal disease, while balloon pulmonary angioplasty and targeted medical therapy can help selected inoperable or residual disease.

Cor pulmonale and chronic right ventricular failure

Cor pulmonale is right ventricular structural or functional change caused by lung disease, pulmonary vascular disease, or chronic hypoxemia rather than primary left heart disease. In COPD, emphysema destroys vascular bed area while hypoxemia drives pulmonary vasoconstriction. In interstitial lung disease, fibrosis reduces capillary cross-sectional area and worsens hypoxemia. In OSA and obesity hypoventilation, repeated nocturnal hypoxemia and hypercapnia increase pulmonary vascular tone. Over time, the right ventricle faces sustained afterload and moves from adaptive hypertrophy to dilation and failure.

Management is cause-directed. Long-term oxygen therapy improves survival in selected COPD patients with severe chronic resting hypoxemia; it also reduces hypoxic pulmonary vasoconstriction. Diuretics relieve edema and hepatic congestion but can reduce RV preload too much, worsening output. Bronchodilators, smoking cessation, pulmonary rehabilitation, vaccination, treatment of sleep-disordered breathing, and management of exacerbations reduce pulmonary stress. PAH-specific vasodilators are not routine therapy for ordinary COPD-related pulmonary hypertension and may worsen ventilation-perfusion mismatch in some patients.

Clinical clue	Likely implication	Common pitfall
Loud P2 plus exertional syncope	Pulmonary hypertension with limited cardiac output reserve.	Calling it anxiety or asthma without checking RV/pulmonary pressure clues.
COPD with JVD, edema, hepatomegaly	Cor pulmonale or mixed left/right heart disease.	Giving large diuretic doses without watching renal function, bicarbonate, BP, and preload.
Persistent dyspnea after treated PE	CTEPH, recurrent PE, deconditioning, lung disease, anemia, or heart disease.	Assuming anticoagulation always restores baseline symptoms.
Hypoxemia worse than expected from CXR	PE, pulmonary vascular disease, shunt, early ILD, or measurement artifact.	Waiting for classic S1Q3T3 or hemoptysis before testing.

High-yield exam clues

- Sudden dyspnea plus pleuritic chest pain plus tachycardia after surgery or immobilization should trigger PE probability assessment.
- Hypotension with suspected PE means high-risk PE until proven otherwise; bedside echo and reperfusion planning may outrank routine D-dimer logic.
- Positive D-dimer does not diagnose PE; infection, cancer, age, pregnancy, trauma, surgery, and inflammation can elevate it.
- Normal oxygen saturation does not exclude PE, especially in young or early presentations.
- Right heart catheterization is the confirmation test for pulmonary hypertension; echocardiography is a screening and risk tool.
- Group 2 pulmonary hypertension from left heart disease is common; do not reflexively treat all pulmonary hypertension with PAH vasodilators.
- CTEPH is a treatable cause of pulmonary hypertension and should be considered with persistent dyspnea after PE.

Common pitfalls

Pitfall	Why it is wrong	Better reasoning
Ordering D-dimer in a high-probability PE patient and stopping when it is negative.	D-dimer is a rule-out tool in low/intermediate probability settings, not a high-risk dismissal tool.	Use imaging or urgent unstable-patient pathway when probability is high.
Waiting for hemoptysis, pleuritic pain, and S1Q3T3 before considering PE.	Most patients lack the full classic triad; ECG can be nonspecific.	Use risk factors, vital signs, oxygenation, and probability tools.
Calling every RV strain PE.	RV strain can come from pulmonary hypertension, RV infarct, severe lung disease, ARDS, or chronic disease.	Integrate history, imaging, biomarkers, and time course.
Treating pulmonary hypertension before identifying group.	Group 2 and group 3 disease are common and may be harmed by inappropriate vasodilator use.	Classify with echo, lung/left heart evaluation, V/Q where appropriate, and right heart catheterization.
Over-diuresing cor pulmonale.	RV output can be preload-sensitive; excessive diuresis may worsen hypotension and kidney function.	Aim for decongestion while monitoring BP, renal function, bicarbonate, and symptoms.

Practice questions

Question 1. A 63-year-old man is 5 days after hip surgery and develops sudden dyspnea, pleuritic chest pain, heart rate 118/min, and oxygen saturation 91%. Blood pressure is normal. What is the most appropriate diagnostic direction?

Answer explanation. This patient has substantial PE probability because of recent surgery, tachycardia, hypoxemia, and sudden pleuritic symptoms. PERC is not appropriate. If stable and no contraindication exists, CTPA is usually the next major diagnostic test; D-dimer is less helpful because probability is not low and postoperative D-dimer is often positive.

Question 2. A 28-year-old woman with no VTE history has brief pleuritic chest pain. HR is 76/min, oxygen saturation is 99%, no hemoptysis, no estrogen use, no unilateral leg swelling, no recent surgery/trauma, and the clinician believes PE probability is low. What tool can prevent unnecessary testing?

Answer explanation. PERC can be used only in low-risk patients. If all criteria are negative, PE can be ruled out without D-dimer or imaging in the correct context.

Question 3. A patient with confirmed PE is normotensive but has RV dilation on echo and elevated troponin. What risk category does this suggest?

Answer explanation. Intermediate-risk/submassive PE. Treatment is anticoagulation and close monitoring, with rescue reperfusion if deterioration occurs rather than routine thrombolysis for everyone.

Question 4. A patient has exertional syncope, loud P2, RV enlargement on echo, and suspected pulmonary hypertension. What test confirms the diagnosis and separates pre-capillary from post-capillary physiology?

Answer explanation. Right heart catheterization. Echocardiography estimates probability and evaluates RV/valves, but catheterization confirms pressures, wedge pressure, cardiac output, and pulmonary vascular resistance.

Question 5. A COPD patient with chronic hypoxemia develops JVD, edema, hepatomegaly, and loud P2. What is the unifying diagnosis and management principle?

Answer explanation. Cor pulmonale. Treat chronic hypoxemia and underlying lung disease, use diuretics carefully for congestion, and avoid assuming all pulmonary hypertension should receive PAH vasodilators.

Graph RAG extraction map

Node	Important outgoing relationships
Pulmonary embolism	Caused by venous thromboembolism; increases pulmonary vascular resistance; can cause RV strain, hypoxemia, pleuritic pain, syncope, obstructive shock; diagnosed by probability-based pathway; treated with anticoagulation and sometimes reperfusion.
DVT	Risk organized by Virchow triad; proximal DVT embolizes to pulmonary arteries; diagnosed by compression ultrasound; treated with anticoagulation or selected surveillance.
D-dimer	Fibrin degradation marker; rules out VTE only when pretest probability is low/intermediate; falsely elevated by age, infection, inflammation, trauma, surgery, pregnancy, malignancy.
Right ventricle	Fails acutely when afterload rises in massive PE; remodels chronically in pulmonary hypertension; preload-sensitive during cor pulmonale decompensation.
Pulmonary hypertension	Classified into WHO groups 1-5; screened by echo; confirmed by right heart catheterization; causes dyspnea, loud P2, syncope, RV failure.
CTEPH	Chronic organized thromboembolic obstruction; follows prior PE or occult emboli; screened with V/Q; potentially treatable with endarterectomy, angioplasty, targeted therapy, anticoagulation.
Cor pulmonale	RV dysfunction from lung disease/hypoxemia; caused by COPD, ILD, OSA/OHS, CTEPH; managed with oxygen when indicated, lung disease treatment, careful diuresis.

Selected references used for medical cross-checking

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